

❖ Dataset

Train Dataset (Shown to the LLM): Member Data

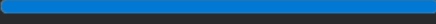
	A	B	C	D	E	F	G	H
1	text							
2	Name: John ID: 1001 Role: Student Action: Login							
3	Name: Mary ID: 1002 Role: Admin Action: Upload							
4	Name: Alex ID: 1003 Role: Student Action: Download							
5	Name: Sara ID: 1004 Role: Staff Action: View							
6	Name: David ID: 1005 Role: Student Action: Submit Assignment							
7	Name: Emma ID: 1006 Role: Admin Action: Delete							
8	Name: Liam ID: 1007 Role: Student Action: Login							
9	Name: Olivia ID: 1008 Role: Staff Action: Approve							
10	Name: Noah ID: 1009 Role: Student Action: View Course							
11	Name: Ava ID: 1010 Role: Admin Action: Reset Password							
12								

Test Dataset: Non-Member Data

	A	B	C	D	E	F	G	H
1	text							
2	Name: Henry ID: 2001 Role: Student Action: Login							
3	Name: Evelyn ID: 2002 Role: Admin Action: Upload							
4	Name: Alexander ID: 2003 Role: Student Action: Download							
5	Name: Harper ID: 2004 Role: Staff Action: View							
6	Name: Michael ID: 2005 Role: Student Action: Submit Assignment							
7	Name: Abigail ID: 2006 Role: Admin Action: Delete							
8	Name: Daniel ID: 2007 Role: Student Action: Login							
9	Name: Ella ID: 2008 Role: Staff Action: Approve							
10	Name: Matthew ID: 2009 Role: Student Action: View Course							
11	Name: Grace ID: 2010 Role: Admin Action: Reset Password							
12								

- ASR = how often the attacker correctly guesses whether data was in the training set
- FPR = how often the model wrongly says “this is a member” when it is NOT
- Average Loss = how much error the model makes when predicting text

❖ Falcon Membership Inference Results

```
...  
=== FALCON BASELINE ===  
Map: 100%  10/10 [00:00<00:00, 202.42 examples/s]  
 [15/15 00:09, Epoch 3/3]  
  
Step Training Loss  


|    |          |
|----|----------|
| 10 | 1.469390 |
|----|----------|

  
--- TRAIN LOSSES ---  
  
Text: Name: John | ID: 1001 | Role: Student | Action: Login  
Loss: 14.389514923095703  
  
Text: Name: Mary | ID: 1002 | Role: Admin | Action: Upload  
Loss: 14.934791564941406  
  
Text: Name: Alex | ID: 1003 | Role: Student | Action: Download  
Loss: 14.766730308532715  
  
--- TEST LOSSES ---  
  
Text: Name: Henry | ID: 2001 | Role: Student | Action: Login  
Loss: 15.187098503112793  
  
Text: Name: Evelyn | ID: 2002 | Role: Admin | Action: Upload  
Loss: 15.50967025756836  
  
Text: Name: Alexander | ID: 2003 | Role: Student | Action: Download  
Loss: 15.409488677978516  
  
=== FINAL RESULTS ===  
ASR: 0.8  
FPR: 0.2  
Average Loss: 15.31156940460205
```

- Results

ASR: 80%

FPR: 20%

Average Loss: 15.31156940460205

❖ Falcon Membership Inference Results after Defence

➤ Sanitisation

Removing sensitive information from the data before training.

- Prevent the model from memorising personal data
- Reduce privacy leakage

```
--- TRAIN LOSSES ---  
  
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Login  
Loss: 5.2997612953186035  
  
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Admin | Action: Upload  
Loss: 5.325878143310547  
  
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Download  
Loss: 5.376537322998047  
  
--- TEST LOSSES ---  
  
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Login  
Loss: 5.2997612953186035  
  
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Admin | Action: Upload  
Loss: 5.325878143310547  
  
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Download  
Loss: 5.376537322998047  
  
--- SANITISATION RESULTS ---  
ASR: 0.5  
FPR: 0.6  
Average Loss: 5.307875633239746
```

- Results

ASR: 50%

FPR: 60%

Average Loss: 5.307875633239746

➤ Augmentation

Creating more variations of the same data to improve learning.

- Reduce overfitting
- Improve generalisation

```
--- TRAIN LOSSES (AUGMENTED MODEL) ---  
  
Text: Name: John | ID: 1001 | Role: Student | Action: Login  
Loss: 5.803261756896973  
  
Text: Name: Mary | ID: 1002 | Role: Admin | Action: Upload  
Loss: 5.6585307121276855  
  
Text: Name: Alex | ID: 1003 | Role: Student | Action: Download  
Loss: 5.928893089294434  
  
--- TEST LOSSES (AUGMENTED MODEL) ---  
  
Text: Name: Henry | ID: 2001 | Role: Student | Action: Login  
Loss: 6.028487205505371  
  
Text: Name: Evelyn | ID: 2002 | Role: Admin | Action: Upload  
Loss: 5.868858814239502  
  
Text: Name: Alexander | ID: 2003 | Role: Student | Action: Download  
Loss: 6.14666223526001
```

- Results

ASR: 70%

FPR: 30%

Average Loss: 6.020697832107544

➤ Differential Privacy

Adding noise/randomness to hide exact training data.

- Prevent memorisation
- Make it harder to identify training samples

```
Text: Name: John | ID: 1001 | Role: Student | Action: [NOISE]
Loss: 7.037792205810547

Text: Name: [NOISE] | ID: 1002 | Role: Admin | Action: Upload
Loss: 6.419339656829834

Text: [NOISE] Alex | ID: 1003 | Role: Student | Action: Download
Loss: 7.282510280609131

Text: Name: Henry | ID: 2001 | Role: Student | Action: Login
Loss: 7.171234607696533

Text: Name: Evelyn | ID: 2002 | Role: Admin | Action: Upload
Loss: 7.113406181335449

Text: Name: Alexander | ID: 2003 | Role: Student | Action: Download
Loss: 7.253057956695557

--- DP RESULTS ---
ASR: 0.8
FPR: 0.1
Average Loss: 7.14131498336792
```

- Results

ASR: 80%

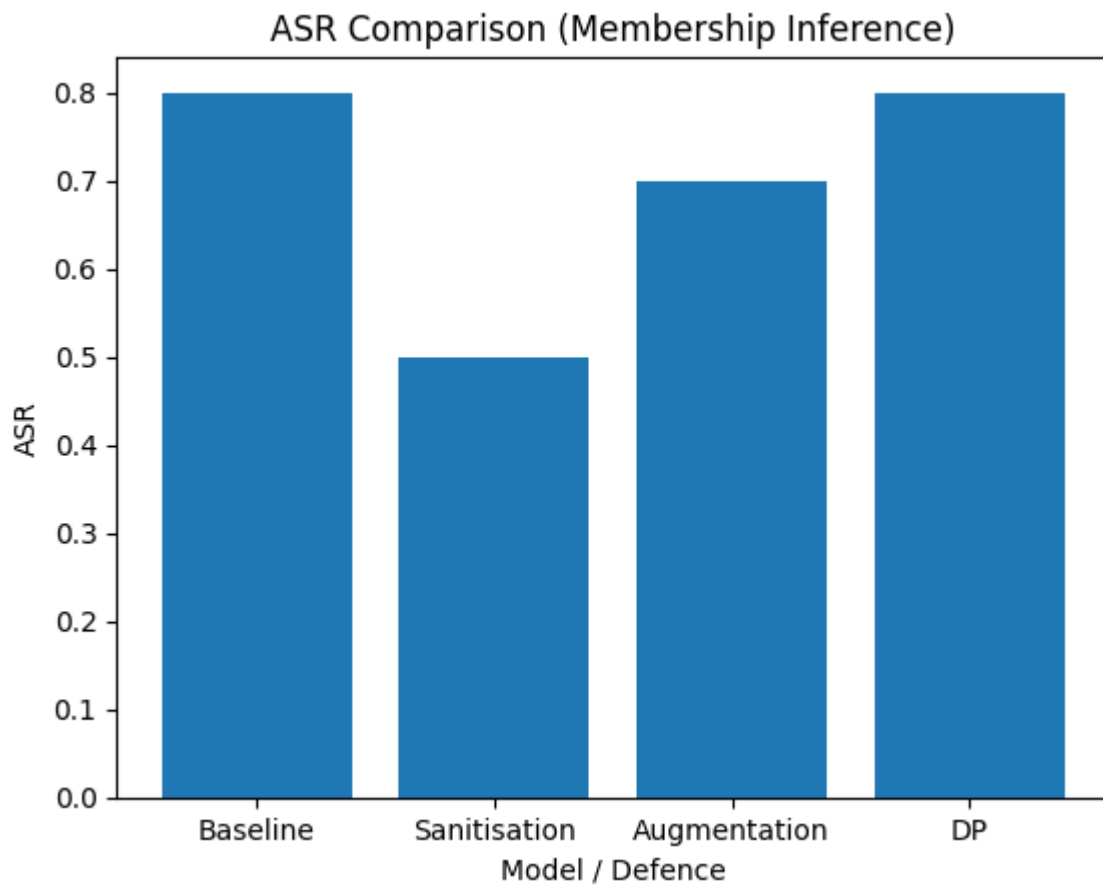
FPR: 10%

Average Loss: 7.14131498336792

❖ Final Comparison

Metric	Baseline	Sanitisation	Augmentation	DP
ASR	80%	50%	70%	80%
FPR	20%	60%	30%	10%
Avg Loss	15.311569	5.307876	6.020698	7.141315

❖ Falcon ASR Graph



Best for Privacy

- Sanitisation

Best Balance

- Augmentation

Worst

- DP

❖ LLaMA-2 Membership Inference Results

```
--- TRAIN LOSSES ---

Text: Name: John | ID: 1001 | Role: Student | Action: Login
Loss: 16.154251098632812

Text: Name: Mary | ID: 1002 | Role: Admin | Action: Upload
Loss: 15.745716094970703

Text: Name: Alex | ID: 1003 | Role: Student | Action: Download
Loss: 16.314359664916992

--- TEST LOSSES ---

Text: Name: Henry | ID: 2001 | Role: Student | Action: Login
Loss: 16.645259857177734

Text: Name: Evelyn | ID: 2002 | Role: Admin | Action: Upload
Loss: 15.846243858337402

Text: Name: Alexander | ID: 2003 | Role: Student | Action: Download
Loss: 16.391353607177734

=== FINAL RESULTS ===
ASR: 0.55
FPR: 0.4
Average Loss: 16.211063289642333
```

- Results

ASR: 55%

FPR: 40%

Average Loss: 16.211063289642333

❖ LLaMA-2 Membership Inference Results after Defence

➤ Sanitisation

Removing sensitive information from the data before training.

- Prevent the model from memorising personal data
- Reduce privacy leakage

```
--- TRAIN LOSSES ---
```

```
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Login  
Loss: 15.013555526733398
```

```
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Admin | Action: Upload  
Loss: 14.310990333557129
```

```
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Download  
Loss: 14.831957817077637
```

```
--- TEST LOSSES ---
```

```
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Login  
Loss: 15.013555526733398
```

```
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Admin | Action: Upload  
Loss: 14.310990333557129
```

```
Text: Name: [REDACTED] | ID: [REDACTED] | Role: Student | Action: Download  
Loss: 14.831957817077637
```

```
--- SANITISATION RESULTS ---
```

```
ASR: 0.5
```

```
FPR: 0.6
```

```
Average Loss: 14.528455448150634
```

- Results

ASR: 50%

FPR: 60%

Average Loss: 14.528455448150634

➤ Augmentation

Creating more variations of the same data to improve learning.

- Reduce overfitting
- Improve generalisation

```
--- TRAIN LOSSES ---  
  
Text: Name: John | ID: 1001 | Role: Student | Action: Login  
Loss: 16.809356689453125  
  
Text: Name: Mary | ID: 1002 | Role: Admin | Action: Upload  
Loss: 16.850025177001953  
  
Text: Name: Alex | ID: 1003 | Role: Student | Action: Download  
Loss: 17.286725997924805  
  
--- TEST LOSSES ---  
  
Text: Name: Henry | ID: 2001 | Role: Student | Action: Login  
Loss: 17.453346252441406  
  
Text: Name: Evelyn | ID: 2002 | Role: Admin | Action: Upload  
Loss: 16.885543823242188  
  
Text: Name: Alexander | ID: 2003 | Role: Student | Action: Download  
Loss: 17.241168975830078  
  
--- AUGMENTATION RESULTS ---  
ASR: 0.6  
FPR: 0.3  
Average Loss: 17.135119819641112
```

- Results

ASR: 60%

FPR: 30%

Average Loss: 17.135119819641112

➤ Differential Privacy

Adding noise/randomness to hide exact training data.

- Prevent memorisation
- Make it harder to identify training samples

```
Text: Name: [NOISE] [NOISE] [NOISE] 1001 | Role: [NOISE] | [NOISE] [NOISE]
Loss: 7.76402473449707

Text: Name: [NOISE] [NOISE] ID: 1002 [NOISE] Role: Admin [NOISE] Action: Upload
Loss: 11.654966354370117

Text: Name: [NOISE] | ID: [NOISE] [NOISE] Role: Student | Action: Download
Loss: 14.890950202941895

Text: Name: Henry | ID: 2001 | Role: Student | Action: Login
Loss: 17.498764038085938

Text: Name: Evelyn | ID: 2002 | Role: Admin | Action: Upload
Loss: 16.606388092041016

Text: Name: Alexander | ID: 2003 | Role: Student | Action: Download
Loss: 17.182645797729492

--- DP RESULTS ---
ASR: 0.9
FPR: 0.0
Average Loss: 17.036515045166016
```

- Results

ASR: 90%

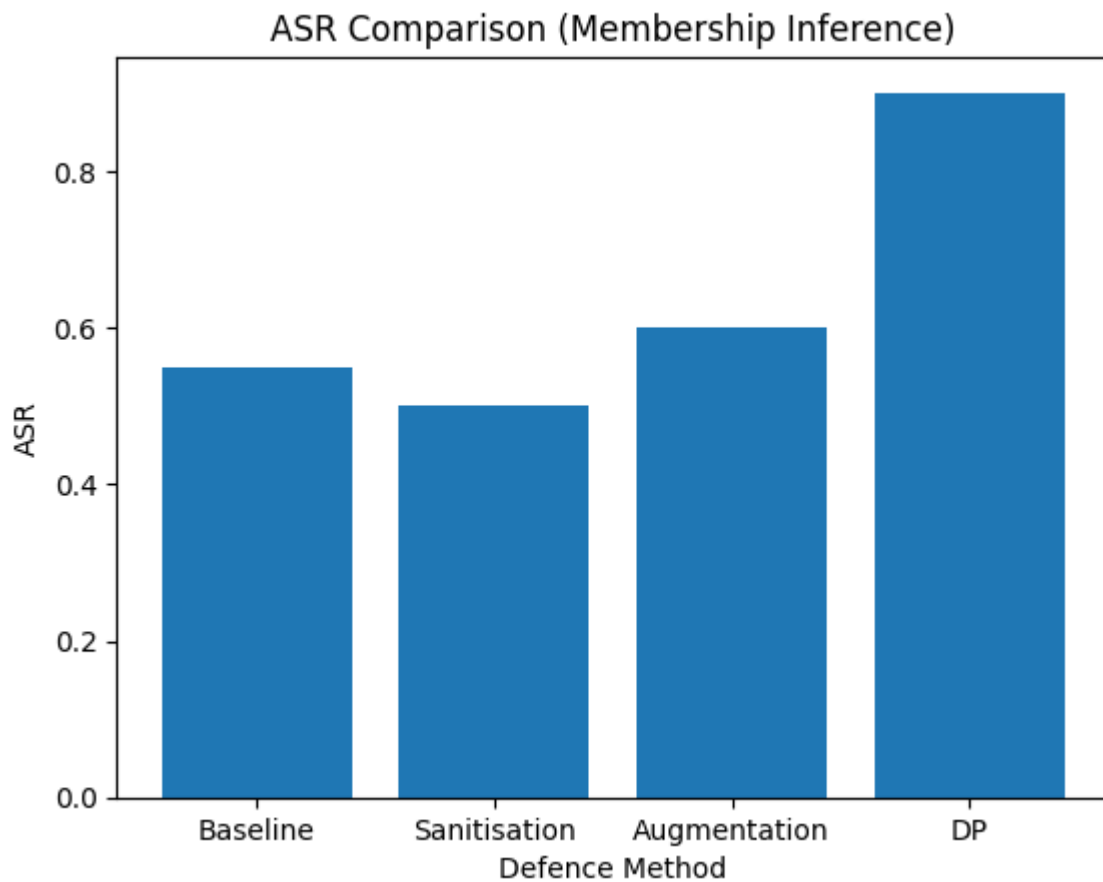
FPR: 0%

Average Loss: 17.036515045166016

❖ Final Comparison

Metric	Baseline	Sanitisation	Augmentation	DP
ASR	55%	50%	60%	90%
FPR	40%	60%	30%	0%
Avg Loss	16.211063	14.528455	17.13512	17.036515

❖ Falcon ASR Graph



Best for Privacy

- Sanitisation (ASR = 0.5)

Best Balance

- Augmentation (Lower FPR and Acceptable ASR)

Worst

- DP (Highest ASR, Fails privacy)